React Mastery — Beginner to Architect

A file-based technical textbook. Each lesson is a Markdown file with the same 15-part structure.

Start here

1. Read ROADMAP.md for the plan, versions, and dependency order.
2. Track your work in PROGRESS.md.
3. Begin at 00 / 01 — How the browser works.

How to study a lesson

1. Read metadata and objectives; check prerequisites.
2. Read sections 2–4 (concept, mechanics, walkthrough). When a lesson says **Predict**, write your prediction down *before* reading on.
3. Run the examples (see below). Change them and break them.
4. Do the exercises (section 11) **without opening the answers** (section 12).
5. Answer the interview questions out loud, then compare.
6. Tick the knowledge checkpoint in PROGRESS.md only if you can do every item without notes.

Running examples

Most examples are small modules. Use a Vite + TypeScript sandbox:

```
npm create vite@latest sandbox -- --template react-ts
cd sandbox && npm install && npm run dev
```

Phase 0 examples run in the browser console or with Node (`node file.mjs`).

Lesson sections

1 Metadata · 2 Concept · 3 Internal mechanics · 4 Execution walkthrough · 5 Code · 6 Visuals · 7 Alternatives & trade-offs · 8 Best practices / anti-patterns · 9 Real-world application · 10 Debugging · 11 Exercises (A–F) · 12 Answers · 13 Interview questions · 14 Resources · 15 Knowledge checkpoint

Honesty conventions

- **Public guarantee** = documented behavior you may rely on. **Teaching model** = simplification. **Implementation detail** = how React currently works internally; it can change.
- External URLs were checked for HTTP 200 on 2026-10-22 (about 920 returned 200); the few still marked *(unverified)* were blocked or unreachable for automated checks. A 200 does not prove the page content matches its description.

- Status in PROGRESS.md is one of: Planned → Drafted → Validated → Completed.

What's here

- 12 phase directories (`00` – `11`) with 265 lessons + phase assessments, 11 projects in `projects/`. Each `X.md` has a matching `X.pdf` next to it.
- REVIEW-NOTES.md lists the version-sensitive claims that need checking against current docs. Treat the curriculum as a strong first draft, not a verified reference.
- Start with Phase 0 (or skip to the phase matching your level); take each phase assessment before advancing (≥80% overall, ≥70% per category).
- Suggested project order is in ROADMAP.md §4.

Rebuilding the PDFs

From `../tools`: `npm install` then `node md2pdf.mjs ../react-mastery` (needs Google Chrome; converts only changed files; `--force` rebuilds all). `node linkcheck.mjs` verifies relative links.